

A7672S-FASE breakout (without GNSS) — ~₹1,300

Yeh LTE **Cat-1** module hai, aur tumhare use case ke liye theek isi ka level chahiye. Bands: **B1/B3/B5/B8** + TDD **B34/B38/B39/B40/B41**, saath mein 2G fallback. India ke liye critical bands **B3, B8, B40** teeno cover hain — Jio, Airtel, Vi sab pe chalega.

Milega: [KTRON ₹1,292](#) (GST included), [Robu](#) (abhi backorder ~20 din), [Valetron](#), [Graylogix](#).

Comparison — kyun yahi

Module	Price	Speed	Verdict
A7672S-FASE	~₹1,300	Cat-1, 10 Mbps	✓ Yahi lo — India bands confirmed, kam power, AT commands SIM7600-compatible
A7670C	<u>~₹899</u>	Cat-1	⚠ ₹400 sasta lekin "C" = China variant. Band risk ke liye ₹400 bachana bewakoofi hai
SIM7600E-H	<u>₹1,900–3,500</u>	Cat-4, 150 Mbps	✗ Tum 200 byte ka JSON bhej rahe ho. 150 Mbps ka koi matlab nahi, aur 2A current bursts
VVM501 (ESP32+4G ek board)	<u>~₹2,400</u>	Cat-1	💎 Wiring bachtai hai, Made in India — lekin ₹1,100 premium aur unke board pe lock
SIM800L	~₹250	2G	✗ Kabhi mat lena. India mein 2G band ho raha hai

GNSS wala variant mat lo (₹200–300 zyada) — parking lock hilta hi nahi, GPS ka koi kaam nahi.

Teen gotchas jo tumhara time barbaad karenge

- 1. Power supply — yahi #1 wajah hai jo log fail hote hain.** LTE transmit pe ~2A ke spikes maarta hai 3.4–4.2V pe. ESP32 ke 3.3V regulator se mat chalo, module baar-baar reboot hoga aur tum firmware debug karte rahoge. Alag buck converter (MP1584/XL4015, 2A+) lagao aur module ke VCC pe **1000–2200 µF ka capacitor** daalo. Yeh non-negotiable hai.
- 2. Jio ki 5G SIM A7672S pe kaam nahi karti** (search results mein specifically likha hai). Ya toh purani 4G-only Jio SIM lo, ya **Airtel** use karo. Testing mein pehle Airtel try karna, kam sirdard.
- 3. Antenna zaroor lagao.** Bina antenna ke module register hi nahi hoga — ₹100–150 ki u.FL LTE antenna, aur usse metal chassis ke bahar rakhna (tumhara steel hub RF block kar dega).

SIM plan

Jio IoT SIM ₹19–99/month mein 1GB, ya Airtel M2M. Tumhara device din mein shayad 1–2 MB bhejega, toh sabse chhota plan bhi bohot zyada hai. Saal ka ~₹600–1,200.

Firmware side

Achhi baat: **A7672S ke AT commands SIM7600 series se compatible hain**, aur MQTT built-in hai. Matlab jo firmware maine likha hai, usme sirf transport layer badalna hai — `WiFiClientSecure` ki jagah `TinyGSM` library (`TinyGsmClientSecure`), baaki MQTT logic, topics, motion control sab waisa ka waisa chalega. Bolo toh woh port kar dun.

Ek line yaad dila dun: yeh ₹1,300 module + ₹150 antenna + ₹200 power circuit = **~₹1,650**, jo tumhare ₹2–3k BOM ko tod deta hai. Isliye 4G ko **commercial/public SKU** ke liye rakho jahan mains power bhi hoga; residential launch BLE + phone bridge pe hi rahe. Abhi ke liye ek piece prototype ke liye lo — poore product ke liye nahi.